

01 Guilty or Not?

A jury must decide whether a defendant is guilty. Map the hypothesis-testing framework onto the courtroom:

- a. What plays the role of H_0 ? Of H_1 ?
- b. What is a Type I error in this context? A Type II error?
- c. What is the "significance level" α analogous to?
- d. In a **criminal** trial, which error is considered worse? What about in a **medical screening** for a serious disease? How does this affect the choice of α ?

$$\frac{1}{0} \quad \frac{1}{6}$$

H_0 : Skuldner eh
vs

H_1 : Skuldur,

P-value

	H_0 skulur	H_0 skulur
reject H_0	Type I Error	Correct
do not reject H_0	Correct	Type II Error

100 95%
 → 1 skulur
 99%

50
 10
 5
 0.05
 100

95%
 5
 0.05

0.05
 0.05

02 The Suspicious Coin

You flip a coin 100 times and observe 60 heads.

- a. State H_0 and H_1 (two-sided).
- b. Compute the z -statistic.
- c. Find the p -value. At $\alpha = 0.05$, do you reject? At $\alpha = 0.01$?
- d. Build a 95% CI for p . Verify that your test decision matches whether $p_0 = 0.5$ lies inside the CI.

$$H_0: p = 0.6$$

$$H_1: p \neq 0.6$$

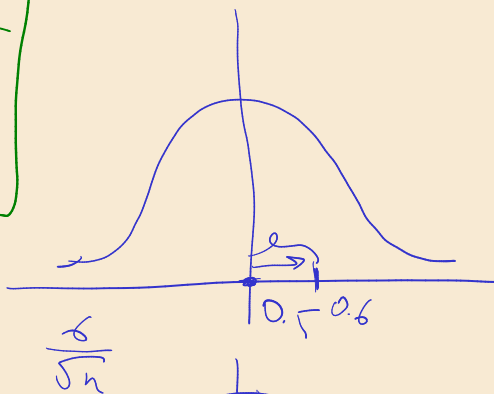
○ H H H ... H

$$H_0: p = 0.5$$

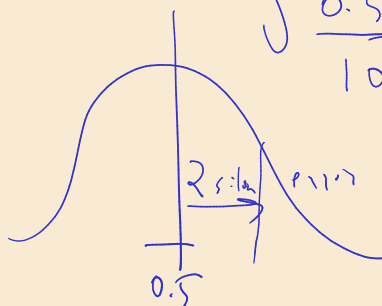
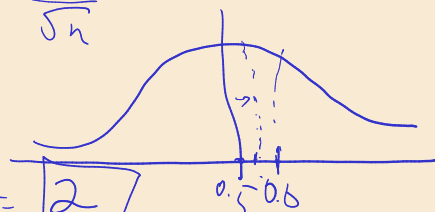
$$H_1: p \neq 0.5$$

Z-statistic

$$SE = \sqrt{\frac{p_0(1-p_0)}{n}}$$

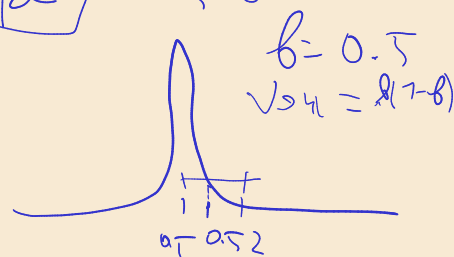


$$\frac{0.6 - 0.5}{\sqrt{\frac{0.5^2}{100}}} = \frac{0.1}{0.05} = 2$$



6, 6

Ⓜ



$$b = 0.5$$

$$\sqrt{0.5(1-0.5)} = 0.5$$

03 p-Value Misconceptions

A study reports $p = 0.03$. For **each** statement below, say TRUE or FALSE and briefly explain.

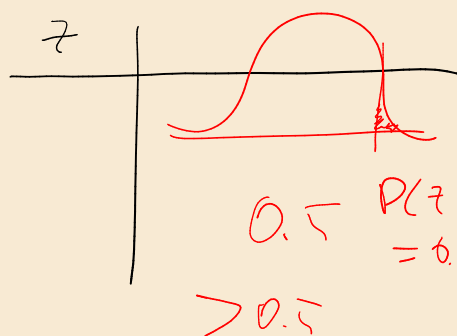
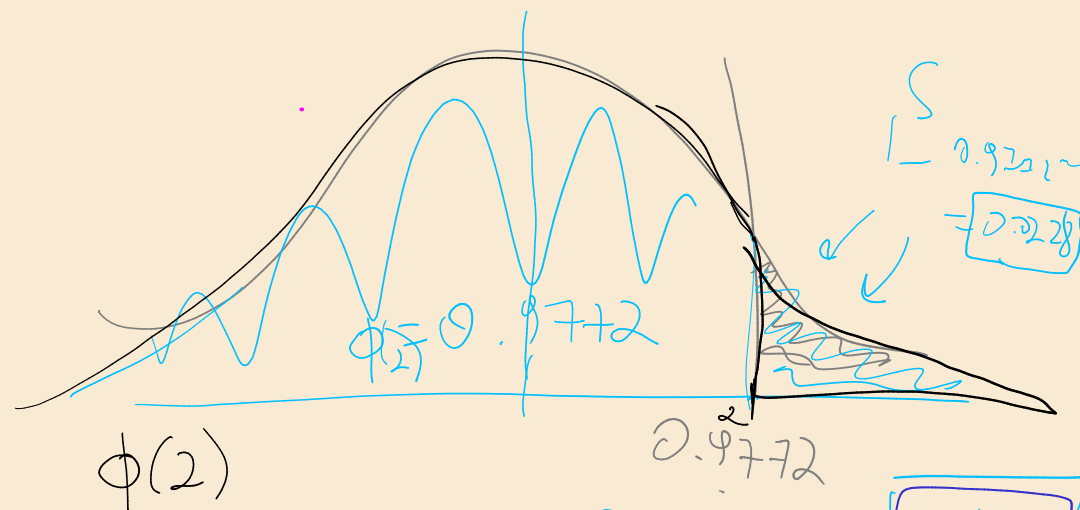
- a. "There is a 3% probability that H_0 is true."
- b. "If we reject H_0 , there is a 3% chance we made an error."
- c. "If we repeated the study many times, we would get $p < 0.05$ at least 97% of the time."
- d. "The probability of observing data this extreme (or more), assuming H_0 is true, is 0.03."
- e. "The effect is practically important."



$$p\text{-value } P(z \geq 2)$$

$$P(z \leq -2)$$

$$P(|z| \geq 2) = 2 P(z \geq 2)$$



$$p = 2 \cdot 0.0228 = 0.0456$$

2.001	0.9773	0.05
2.052	0.9775	0.01
2.001	0.9775	
z	0.01	0.02
2.0	0.9773	0.9775

$p = 0.5$ $p = 0.5$

04 The p-Hacking Experiment

Simulate the multiple-testing disaster in Python:

- a. Generate 20 independent datasets, each containing two groups of $n = 30$ drawn from **the same** $N(0, 1)$ (so H_0 is true for all 20). Run a two-sample t -test on each pair. How many give $p < 0.05$?
- b. Repeat the entire experiment 1,000 times. In what fraction of repetitions do you find **at least one** significant result ($p < 0.05$)?
- c. Compare with the theoretical answer: $1 - (1 - 0.05)^{20} \approx ?$
- d. Take one of the runs where several tests are "significant." Apply **Bonferroni** and **Benjamini-Hochberg** corrections. How many remain significant after each?

